Clk	PC (Start)	Instruction	IR (Binary)	Opcode	Operation
0	0	R1 <- NSH(R1)	1000001	0	No-Shift (NSH)
1	1	R1 <- R0 + R1	1E+12	1	Add ($5 + 8 = 13$)
2	2	if R0 > R1 then branch to 3	1.111E+15	1111	Branch If Greater (13 > 8)
3	3	R1 <- ZERO(R1)	1001010001	0	Zero Register
4	4	if ZF=1 then branch to 7	1.101E+15	1101	Branch If ZF=1 (ZF is 1)
5	7	R3 <- R7 * R3	1.01011E+12	1	Multiply ($4 \times 6 = 24$)
6	8	R1 <- R5 / R7	1.0111E+12	1	Divide ($36/4 = 9$)
7	9	if R1 > R7 then branch to 5	1.111E+15	1111	Branch If Greater ($9 > 4$)
8	5	R2 <- SHR(R5)	1.01101E+11	0	SHR ($36 = 00100100_2$). R5 $\leftarrow$ $00010010_2 = 18$. CF $\leftarrow$ 0. R2 $\leftarrow$ 18.
9	6	LAD (R0 $\leftarrow$ AC+1)	1.01E+15	1010	INC AC (AC holds R5=36). AC $\leftarrow$ $36 + 1 = 37$. R5 is unaffected.
10	10	R0 $\leftarrow$ R7 $\times$ R3	0001 0101 1101 1000	1	**R0** $\leftarrow$ **R7** $\times$ **R3** $\mathrm{AC} \leftarrow 4 \times 36 \rightarrow 96$ (User specified result). $\mathrm{MQ} \leftarrow 36$. $\mathrm{R0} \leftarrow 96$.
11	11	**R2** $\leftarrow$ **R0 AND R7**	0001 1000 0011 1010	1	**R2** $\leftarrow$ **R0 AND R7** (AND). $\mathrm{AC} \leftarrow \mathrm{R0\ AND\ R7} = 96\ \mathrm{AND}\ 4 = 0$. $\mathrm{MQ} \leftarrow \mathrm{R7} = 4$. $\mathrm{R2} \leftarrow 0$.
12	12	**R5** $\leftarrow$ **ROR(R5)**	0000 1101 0100 0101	0	R5 $\leftarrow$ ROR(R5). $\mathrm{AC} \leftarrow \mathrm{R5} = 36$. $\mathrm{ROR(AC)} = 18$. $\mathrm{R5} \leftarrow 8$.
13	13	R2 $\leftarrow$ AC	0010 0000 0000 0010	10	R2 $\leftarrow$ AC (Store to R2). $\mathrm{AC} = 18$. $\mathrm{R2} \leftarrow 18$.
14	14	**AC** $\leftarrow$ **R0, MQ** $\leftarrow$ **R6**	0011 0001 0111 0000	11	AC $\leftarrow$ R0, MQ $\leftarrow$ R6 (Load). $\mathrm{AC} \leftarrow \mathrm{R0} = 96$. $\mathrm{MQ} \leftarrow \mathrm{R6} = 9$.

Flags (SF, ZF, CF)	R0	R1	R2	R3	R4	R5	R6	R7	Next PC
0, 0, 0	5	8	12	6	16	36	9	4	1
0, 0, 0	5	13	12	6	16	36	9	4	2
0, 0, 0	5	13	12	6	16	36	9	4	3
0, 1, 0	5	0	12	6	16	36	9	4	4
0, 1, 0	5	0	12	6	16	36	9	4	7
0, 0, 0	5	0	12	24	16	36	9	4	8
0, 0, 0	5	9	12	24	16	36	9	4	9
0, 0, 0	5	9	12	24	16	36	9	4	5
0, 0, 0	5	9	18	24	16	36	9	4	6
0, 0, 0	5	9	18	24	16	36	9	4	10
0, 0, 0	96	9	18	24	16	36	9	4	11
0, 1, 0	96	9	0	24	16	36	9	4	12
0, 0, 0	96	9	0	24	16	18	9	4	13
0, 0, 0	96	9	18	24	16	18	9	4	14
0,0, 0	96	9	18	24	16	18	9	4	15